

CLINICAL ARTICLE

Influence of Experimental Lighting Conditions on Digital Shade Matching and Intraoral Scanning Accuracy: A Clinical Study

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ABSTRACT

Objectives: To evaluate the in vivo effect of varying experimental lighting conditions on the digital shade determination and scanning accuracy (trueness and precision) of two intraoral scanners (IOSs).

Material and Methods: The maxillary arches of 10 volunteers were scanned using two IOSs (Trios 4 wireless, 3Shape A/S [TR]; and Primescan, Dentsply Sirona [PR]). Scans were performed under four illuminance levels (0, 500, 1000, and 2500 lx) with scanner-specific color temperature (TR: 5500 K; PR: 4100 K). Digital shade selection was performed at the middle third of the right central incisor against visual consensus by three experts for Vita toothguide Classical (VC) and Vita toothguide 3D-Master (VM) shade guides. Scanning accuracy was assessed by superimposing IOS scans onto reference datasets obtained by digitizing conventional stone models with a laboratory scanner (InEos X5; Dentsply Sirona). Shade matching capability (accuracy and reliability) was also evaluated. Data were statistically analyzed using one-way ANOVA ($\alpha=0.05$).

Results: Ambient lighting significantly influenced digital shade matching and scanning trueness ($p<0.05$). For shade matching accuracy, TR demonstrated the highest values at 1000 lx/5500 K (VC: $93.33\% \pm 14.05\%$; VM: $76.67\% \pm 22.50\%$), while PR performed best at 2500 lx/4100 K (VC: $96.67\% \pm 10.54\%$; VM: $50.00\% \pm 28.33\%$). TR produced significantly lower reliability results at the “no-light” (0 lx) condition ($p<0.05$), whereas PR displayed no significant difference among groups ($p>0.05$). For scanning trueness, TR exhibited the lowest deviation at 1000 lx/5500 K ($104.04 \pm 20.88 \mu\text{m}$), while PR performed best at 2500 lx/4100 K ($100.78 \pm 19.36 \mu\text{m}$).

Conclusions: Ambient lighting conditions significantly affected IOS performance in vivo. The optimal conditions were scanner-dependent: TR performed best under typical examination room lighting (1000 lx, 5500 K), while PR excelled under dental operatory lighting (2500 lx, 4100 K).

Clinical Significance: To optimize digital impressions and esthetic shade matching, clinicians should adjust ambient lighting according to the specific requirements of the intraoral scanner being used.

1 | Introduction

Accurate tooth shade determination is critical for achieving optimal esthetic outcomes, particularly in the anterior region, yet it remains one of the most significant challenges in restorative dentistry [1–3]. With the rapid advancement of digital workflows, intraoral scanners (IOSs) have been integral to contemporary practice [4–6]. Beyond capturing high-resolution topography, these devices acquire true-color digital impressions without the need for opaquing powders, enabling simultaneous shade assessment during the scanning process [7–12].

Recent systematic reviews have demonstrated that IOSs exhibit promising repeatability in shade selection, with agreement rates ranging from 75% to 94%, and may provide more consistent esthetic outcomes than conventional visual shade selection (VSS) methods [13, 14]. However, most studies on IOS-based digital shade matching have been limited to reporting in vitro agreement with shade tabs or to qualitative clinical findings without statistical evaluation [9, 10, 15–18], leaving the clinical relevance of these findings unclear. Whether current IOS-based shade determination consistently meets clinically acceptable outcomes in routine practice remains a subject of ongoing debate.

Several variables influence IOS performance, including scanner-related characteristics [19–22], object properties [23, 24], scanning strategy [25], and ambient lighting conditions [18, 26–32]. Among these, ambient lighting is an unavoidable yet modifiable variable in clinical settings [26, 33]. Previous studies have reported that lighting significantly influences digital shade matching [32, 34–36]. For instance, illuminance levels ranging from 0 to 10,000 lx have been shown to significantly influence shade selection outcomes [18]. Reyes et al. [37] further observed higher IOS shade-matching reliability under natural light (6921 lx) than under a typical clinical setting (1808 lx). However, most previous studies have focused primarily on illuminance, while the role of light source color temperature remains largely unexplored in the context of in vivo IOS-based shade determination.

Evidence suggests that specific light sources may enhance both scanning accuracy and shade selection [26, 28, 32, 33]. A previous in vitro study indicated that both illuminance and color temperature significantly influence performance: Trios 4 wireless (3Shape A/S, Denmark) achieved optimal scanning and shade-matching results at 1000 lx and 5500 K, whereas Primescan (Dentsply Sirona, USA) demonstrated superior performance at 2500 lx and 4100 K [33]. Although our previous study established scanner-specific optimal lighting conditions in vitro, that investigation relied on resin-based shade tabs, whose color range and optical behavior cannot fully reproduce the complex light-scattering properties of the natural enamel-dentin complex [38], thereby limiting the clinical translatability of those findings. The complex realities of the oral cavity may substantially modify the spectral and topographic input received by IOSs. The extent to which the previously identified optimal conditions hold in this complex clinical environment remains unverified. Consequently, evidence-based guidelines for standardizing the dental operator environment are still lacking. Therefore, this in vivo study aimed to investigate the impact of varying ambient lighting conditions on the shade-selection capacity and scanning accuracy of two widely used IOSs. The null hypothesis was

that, under in vivo conditions, varying ambient lighting would not produce statistically significant differences in IOS-based shade matching capability or scanning accuracy.

2 | Material and Methods

The overall study workflow is illustrated in Figure 1.

2.1 | Ethical Approval and Volunteer Selection

This study was conducted in accordance with the guidelines and policies of the ethics committee and the human subjects oversight committee. Ethical approval was obtained from the Institutional Review Board of the local University (approval number: 2022-60). All volunteers provided written informed consent to participate after receiving a detailed explanation of the study protocol.

The inclusion criteria were: (1) age 18–25 years with intact, well-aligned dentition, free from crowding, torsion, or other malocclusions; (2) fully erupted maxillary central incisors with vital pulps and no restorations [39]; and (3) ability to comply with the digital-impression process [40].

The exclusion criteria were: (1) severe intrinsic tooth discoloration, (e.g., nonvital teeth, fluorosis, or tetracycline staining); (2) periodontal or oral mucosal disorders; (3) active smoking; (4) ongoing orthodontic treatment; (5) previous bleaching treatment; (6) systemic diseases; and (7) symptoms of temporomandibular joint disorder [39].

Sample size calculation was performed using G*Power software (v3.1.9.6f; Düsseldorf, Germany). Based on the primary outcome (shade-matching accuracy), with an alpha level of 0.05, power of 0.80, and an effect size derived from preliminary data, a minimum of 10 participants was required to detect significant differences among groups. All participants received dental prophylaxis 1 week before the experiment and were instructed to avoid smoking, colored beverages, and tooth-whitening products thereafter.

2.2 | Digital Impression and Digital Shade Selection

Two confocal-based IOSs were evaluated: Trios 4 wireless (3Shape A/S, Copenhagen, Denmark; TR) and Primescan (Dentsply Sirona, York, PA, USA; PR). All scans were performed in a controlled environment without natural light. The ambient temperature was maintained at $22^{\circ}\text{C} \pm 1^{\circ}\text{C}$, with a relative humidity of 55% and an atmospheric pressure of 760 mmHg, to ensure consistent environmental conditions across all experimental procedures. Based on preliminary findings [33], TR was tested under illuminance levels of 500, 1000, and 2500 lx at a color temperature of 5500 K, whereas PR was tested under the same illuminance levels at 4100 K. The scanner-specific color temperatures applied (TR: 5500 K; PR: 4100 K) were predetermined based on our prior comprehensive laboratory evaluation. In that study, specific color

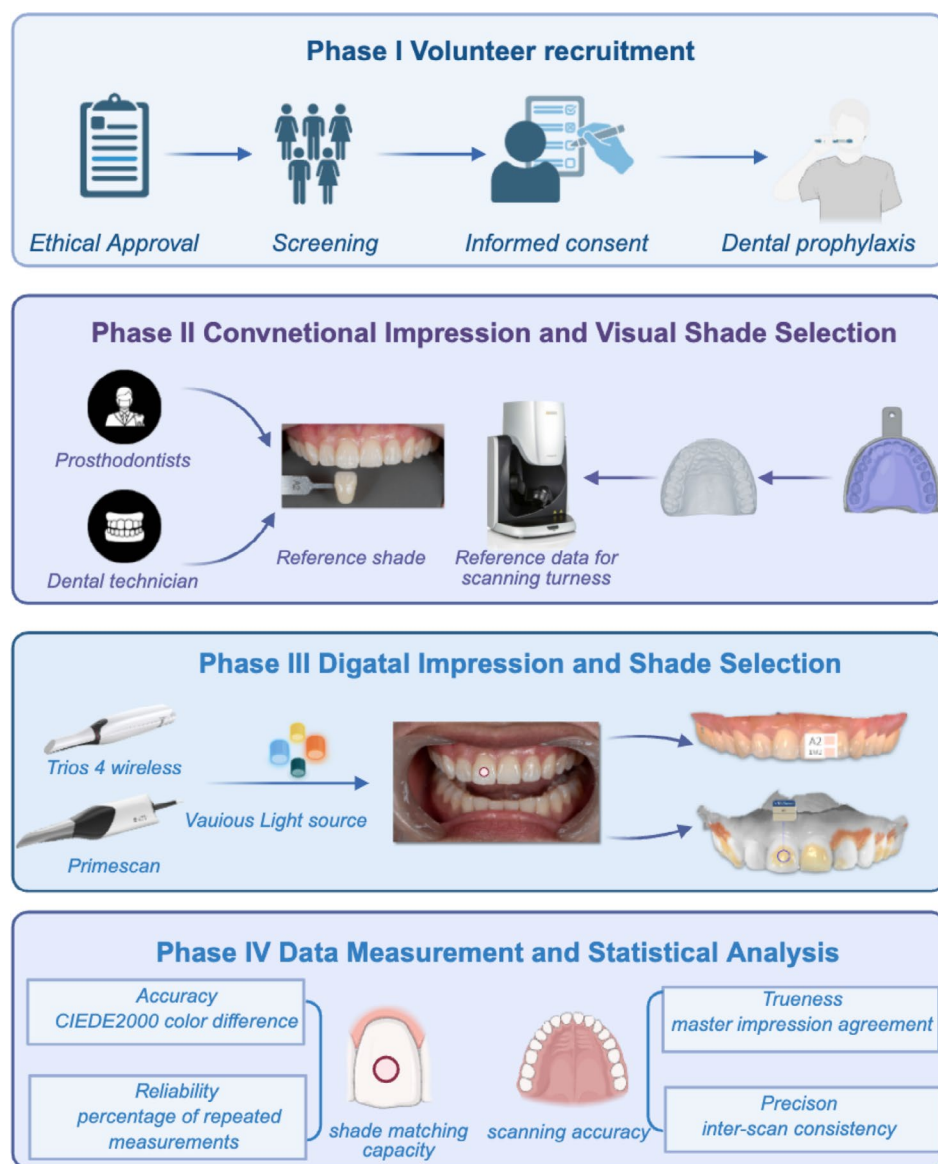


FIGURE 1 | Workflow of the present study.

temperatures consistently yielded optimal color-determination accuracy for each scanner across various illuminance levels [33]. To isolate the clinical effect of illuminance in the present in vivo study, the optimal color temperature for each scanner was maintained as a fixed constant. This sequential design not only allowed us to evaluate the specific impact of illuminance without confounding variables but also minimized the number of repeated intraoral scans required, thereby preventing participant fatigue and physical discomfort. Illuminance and color temperature were verified using a spectral illuminance meter (HPCS-320; Hpopocolor, Hangzhou, China). A 0 lx condition was included as the control to evaluate the maximal influence of ambient lighting deprivation. The experimental light sources were mounted on the scanner head as previously described [33].

Prior to scanning, the IOSs were calibrated according to manufacturer guidelines, including the color-determination module. Conventional maxillary impressions were then taken using

silicone rubber material (Express XT Penta H; 3M ESPE) to fabricate stone casts. The casts were digitized using a high-precision laboratory scanner (InEos X5; Dentsply Sirona) to generate reference STL files for trueness evaluation.

The shade-detection function was then activated, and the scanning strategy followed previously established instructions [15]. The color-picker tool of the software was employed to determine shades in the middle third portion of right central incisor, and results were expressed in color codes corresponding to Vita Toothguide Classical (VC) and Vita Toothguide 3D Master (VM) shade guides (Figure 2). All the scans were conducted by the same experienced prosthodontist (Y.G.) in accordance with each manufacturer's recommended scanning strategy to standardize the operational protocol and minimize variability in scanning distance and angulation. For TR, the scanning angle dynamically ranges from 45° to 90°, and the scanning distance is controlled within the focal depth of approximately 5 mm. For PR, the scanning angle was dynamically maintained between

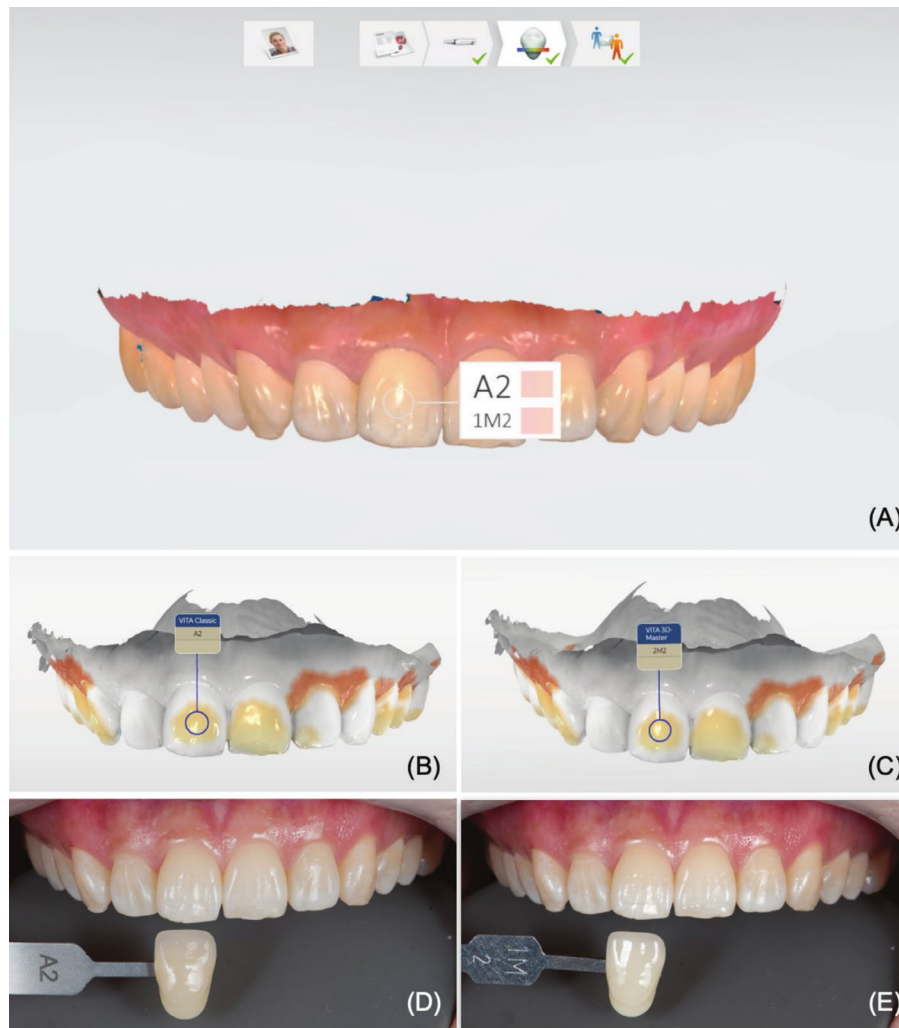


FIGURE 2 | Shade matching outcomes, (A) Trios 4 wireless; (B) Primescan using Vita toothguide classical (VC) shade guides; (C) Primescan using Vita toothguide 3D Master (VM) shade guides; (D) visual shade selection using VC shade guides; (E) visual shade selection using VM shade guides.

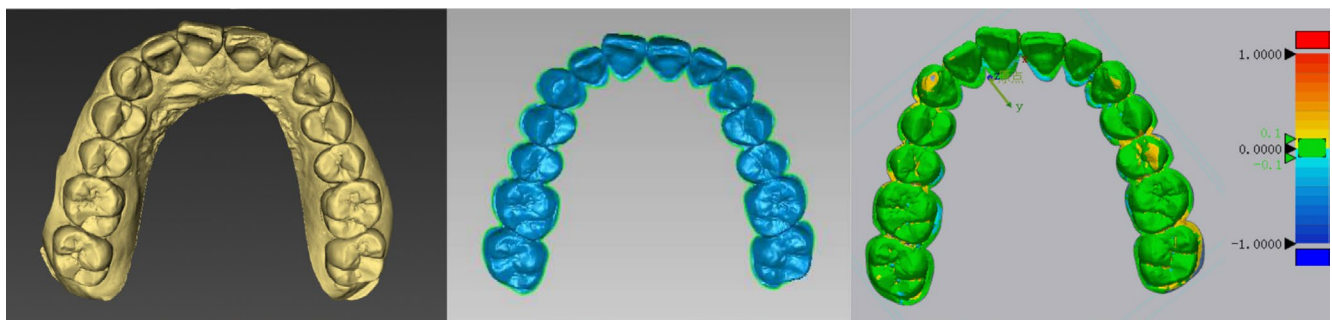


FIGURE 3 | Model trimming and fitting registration.

30° and 60° relative to the tooth surface, and the scanning distance was kept within 2 mm. For each ambient light condition, 3 digital scans were sequentially obtained, with a 15-min interval between each to prevent dehydration of the tooth surface and operator fatigue [18]. All data were exported in the STL format for subsequent assessment. Before study initiation, the operator (Y.G.) underwent training and calibration for the measurement procedure, demonstrating acceptable reliability ($\kappa > 0.80$).

2.3 | Visual Shade Selection (VSS)

Prior to the study, all observers were screened for color vision and color discrimination competence in accordance with International Organization for Standardization (ISO)/Technical Report (TR) 28,642:2016 [41, 42]. Inter-rater and intra-rater reliability were subsequently evaluated using the intraclass correlation coefficient (ICC) analysis. Two experienced prosthodontists

and one dental technician verified the middle third of the upper right central incisor of each volunteer using the VC and VM shade guides, with measurements repeated three times (Figure 2). To minimize visual fatigue, each matching session was limited to 5 s, and a 5-min interval was maintained between shade matches [11]. The most frequently observed shade among all repeated measurements was determined as the “true” tooth color, serving as the reference for assessing shade-selection accuracy [43].

2.4 | Data Measurement

After obtaining the STL data, the impressions in each test group were trimmed to include all dental hard tissues and 1 mm of attached gingiva using qualified software (Geomagic Wrap 2021; 3D Systems, Rock Hill). All data measurements and processing steps were conducted by an independent evaluator. For the trueness measurement, all obtained test data were superimposed with a best-fit algorithm of an inspection software program (Geomagic Control X; 3D Systems, Rock Hill, USA) as recommended by the ISO12836-1:2015 [44]. To measure the precision, the data acquired from repeated scans by each group with each intraoral scanner were paired [45]. The root mean square (RMS) estimates between the registered models and deviation distribution color maps were generated through 3D deviation analysis (Figure 3).

According to ISO 5725-1:2023 [45], trueness and precision describe digital impression accuracy. Trueness reflects the agreement between impression data and the master model, typically assessed by comparing data with high-precision optical scanner references, whereas precision measures consistency across multiple independent scans of the same sample. For shade selection capability, accuracy was calculated as the percentage of digital shade assignments that perfectly matched the VSS consensus reference. Reliability (repeatability) was defined as the percentage of repeated shade measurements that yielded identical outcomes within each specific lighting group [15, 16]. All measurements were performed by a single blinded examiner.

2.5 | Statistical Analysis

Statistical analyses were conducted using SPSS software program (version 25, IBM Corp). Intra-examiner reliability across all measurements was evaluated using the ICC. Normality and homogeneity of variance were assessed using the Shapiro–Wilk and Levene’s tests, respectively. If the tests indicated normal distributions and homoscedasticity, a one-way measures analysis of variance (ANOVA) was performed to investigate the effect of ambient light sources on the shade selection capability and scanning accuracy. Where ANOVA indicated significant differences, post hoc pairwise comparisons were performed using the Tukey test, with the significance level set at $\alpha = 0.05$.

3 | Results

This study included 10 volunteers (6 males and 4 females) with a mean age of 23.9 ± 1.2 years. A total of 240 STL files were

TABLE 1 | Shade matching performance under different ambient light conditions based on VC and VM shade guides using Trios 4 wireless scanner.

	0 lx		500 lx/5500 K		1000 lx/5500 K		2500 lx/5500 K	
	VC	VM	VC	VM	VC	VM	VC	VM
Accuracy (%)	46.67 ± 32.20 ^a	23.33 ± 27.44 ^a	60.00 ± 37.84 ^b	36.67 ± 42.89 ^a	93.33 ± 14.05 ^c	76.67 ± 22.50 ^b	73.33 ± 34.42 ^b	50.00 ± 36.00 ^c
Reliability (%)	66.67 ± 22.22 ^a	63.33 ± 18.92 ^a	86.67 ± 17.21 ^b	80.00 ± 17.21 ^b	93.33 ± 14.05 ^b	83.33 ± 17.57 ^b	83.33 ± 23.57 ^b	80.00 ± 17.21 ^b

Note: Different lowercase letters within each row of the table indicate statistically significant differences between groups using the same shade guides ($p < 0.05$). VC, Vita Toothguide Classical shade guides; VM, Vita Toothguide 3D Master shade guides.

obtained for analysis. Inter-rater and intra-rater ICC values for visual shade selection and measurement demonstrated excellent reliability across all assessments ($\kappa > 0.85$).

3.1 | Shade Matching Capability

Across all experimental groups, shade-selection accuracy ranged from 0 to 100.00%, and shade matching reliability varied between 33.33% and 100.00%.

For the TR scanner, when using the VC shade guide as a reference, shade matching accuracy differed significantly among lighting conditions ($p < 0.05$), with the 1000lx/5500K condition showing the highest accuracy ($p < 0.05$). Shade matching reliability was also significantly inferior in the 0lx group than the other groups ($p < 0.05$). With the VM shade guide as a reference, significant differences were also found for shade-matching accuracy among the tested conditions ($p < 0.05$), and the 0lx condition showed significantly lower accuracy than all other conditions ($p < 0.05$). Overall, the 1000lx/5500K condition yielded the highest accuracy (VC: $93.33\% \pm 14.05\%$; VM: $76.67\% \pm 22.50\%$) and the highest reliability (VC: $93.30\% \pm 14.05\%$; VM: $83.30\% \pm 17.60\%$) (Table 1). Reliability in the 0lx condition was significantly lower than in the other conditions ($p < 0.05$), whereas no significant differences were observed among the remaining conditions (Figure 4).

For PR, using the VC shade guide as a reference, shade matching accuracy differed significantly among lighting conditions ($p < 0.05$), and the percentage of agreement with the reference shade under 2500lx/4100K was significantly higher than under 0lx ($p < 0.05$). Using the VM shade guide as a reference, statistically significant differences were also observed among groups for shade matching accuracy ($p < 0.05$). Under 2500lx/4100K, PR achieved the highest accuracy (VC: $96.67\% \pm 10.54\%$; VM: $50.00\% \pm 28.33\%$) and reliability (VC: $96.67\% \pm 10.54\%$; VM: $83.30\% \pm 17.57\%$) (Table 2). In contrast, shade-selection reliability did not differ significantly among PR lighting conditions ($p > 0.05$) (Figure 4).

3.2 | Scanning Accuracy

Across all experimental groups, scanning trueness ranged from 79.40 to 196.30 μm , and precision ranged from 28.50 to 65.90 μm . For both the TR and PR groups, statistically significant differences were detected in scanning trueness across different

ambient lighting conditions ($p < 0.05$). TR exhibited optimal scanning trueness at 1000lx/5500K ($104.04 \pm 20.88 \mu\text{m}$), while the lowest trueness (highest deviation) was observed in the 0lx condition ($140.91 \pm 19.05 \mu\text{m}$) ($p < 0.05$). PR demonstrated the highest trueness under 2500lx/4100K ($100.78 \pm 19.36 \mu\text{m}$), whereas the lowest trueness was observed under 0lx ($132.51 \pm 16.19 \mu\text{m}$) ($p < 0.05$). No statistically significant differences in scanning precision were observed among groups ($p > 0.05$) (Figure 5).

4 | Discussion

The findings of this study demonstrated that ambient lighting conditions significantly affected both the shade-matching performance and scanning accuracy of the evaluated IOSs, leading to the rejection of the null hypothesis. To our knowledge, this is the first in vivo study to concurrently investigate IOS performance in shade determination and scanning accuracy under systematically varied ambient lighting conditions. Compared to in vitro models [18, 27, 28, 32, 33], clinical research better replicates relevant conditions such as saliva, gingival reflectance, soft-tissue interference, patient-related variables, and the complex light-scattering properties of natural dentition, thereby providing more direct evidence to support clinical decision-making.

Currently, no universally accepted reference standard exists for defining “true” tooth color in vivo. Clinical research typically relies on either instrumental measurements or VSS [7, 8, 11, 12]. Instrumental devices determine tooth color by measuring the reflectance of visible light across different wavelengths and converting the resulting spectral curves into shade codes or CIE $L^*a^*b^*$ values. Despite their high accuracy, these instruments have occasionally shown lower reliability than VSS performed by experienced prosthodontists [46–49]. Given its high repeatability and clinical relevance, consensus-based (modal) VSS conducted by multiple experienced observers is widely recognized as a pragmatic reference standard [9, 15, 50] and was therefore adopted in the present study to assess IOS-based shade matching agreement.

In this study, ambient light significantly influenced the shade selection accuracy of both TR and PR, consistent with previous studies [14, 18, 22]. Moreover, shade assessment accuracy performed under appropriately configured ambient lighting consistently remained within clinically acceptable limits compared to the 0lx (light deprivation) condition; under optimal lighting,

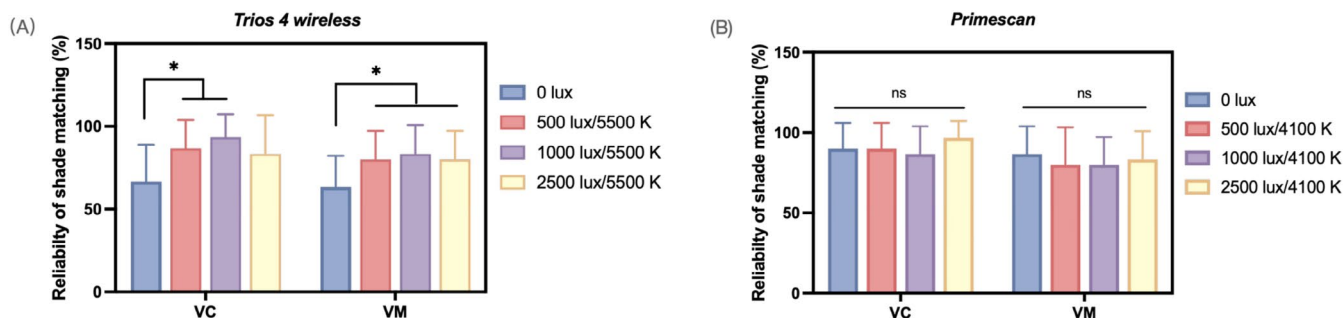


FIGURE 4 | Shade matching reliability under various ambient conditions, (A) Trios 4 wireless; (B) Primescan.

the digital shade assignments reached high agreement with the clinical reference, reaching up to 90%. It has been reported that inaccuracies in IOS-based shade matching may originate from the unpolarized illumination emitted by the device's built-in light-emitting diode light sources [8]. This unpolarized illumination tends to record tooth shades as brighter than their true appearance [1]. Therefore, these findings indicate that appropriate control of ambient lighting can substantially improve IOS-based shade determination. In contrast to a previous study [33], which reported higher shade-matching accuracy with the VM shade system, the present study demonstrated higher shade-matching accuracy when using the VC system. This discrepancy may be attributed to differences in shade distribution and system coverage [49]. In the present cohort, tooth base colors were concentrated primarily within A1 and A2 in the VC system, whereas VM references spanned a broader color space (1M2, 2M1, 2M2, 2L1.5, and 2R1.5), which may influence shade matching outcomes. Clinically, these results suggest that for non-esthetic zones, IOS-based shade selection under appropriate ambient lighting can meet esthetic demands. For esthetic zones or patients with high expectations, IOS outputs should be considered a preliminary reference.

The shade-selection reliability of the IOSs ranged from 70.0% to 93.3%, consistent with previous reports [14–16, 37] and comparable to that of spectrophotometric instruments. In contrast, Ebeid et al. reported lower reliability values when using the VC shade guide [16], which may be attributable to disparities in optical properties between ceramic blocks and natural enamel, as well as the inclusion of a broader range of darker and more chromatic shades (A1, A2, A3, A3.5, A4, B2, B3, C2, C3, and D3). Collectively, these findings suggest that shade-matching reliability may decrease as baseline tooth color becomes darker or more chromatic, underscoring the need for future studies to include more diverse baseline shades.

The present in vivo study also demonstrated that ambient lighting significantly affected scanning trueness, whereas precision remained unaffected. TR achieved the highest scanning accuracy under the 1000lx condition, while PR performed best at 2500lx, successfully translating previous in vitro findings into a clinical context [33]. Evaluating in vivo scanning trueness is inherently challenging due to the absence of an absolute intraoral standard. Consequently, digitizing high-precision conventional impressions or stone casts with an industrial-grade laboratory scanner remains the most pragmatic and widely accepted reference method in current clinical research. Using this established methodology, our results are consistent with those of Koseoglu et al. [51] and Revilla-León et al. [52], who reported optimal scanning accuracy at approximately 1000lx and the poorest performance under 0lx conditions. Notably, unlike in vitro studies, the 0lx condition in the present in vivo study exceeded the clinically acceptable threshold (120µm). This underscores the clinical risk of relying solely on a scanner's built-in light, particularly in restricted-access areas like the distal surfaces of maxillary molars.

Several limitations should be acknowledged. First, although the sample size of 10 participants was statistically justified via power analysis to detect primary outcome differences, it remains relatively small. As a pilot-scale clinical investigation, all

TABLE 2 | Shade matching performance under different ambient light conditions based on VC and VM shade guides using Primescan.

	0 lx		500 lx/4100 K		1000 lx/4100 K		2500 lx/4100 K	
	VC	VM	VC	VM	VC	VM	VC	VM
Accuracy (%)	53.33 ± 47.66 ^a	23.33 ± 41.72 ^a	63.33 ± 36.68 ^a	46.67 ± 35.83 ^b	76.67 ± 27.44 ^b	50.00 ± 28.33 ^b	96.67 ± 10.54 ^b	66.67 ± 31.43 ^c
Reliability (%)	90.00 ± 16.10	86.67 ± 17.21	90.00 ± 16.10	80.00 ± 23.31	86.67 ± 17.21	80.00 ± 17.21	96.67 ± 10.54	83.33 ± 17.57

Note: Different lowercase letters within each row of the table indicate statistically significant differences between groups using the same shade guides ($p < 0.05$). VC, Vita Toothguide Classical shade guides; VM, Vita Toothguide 3D Master shade guides.

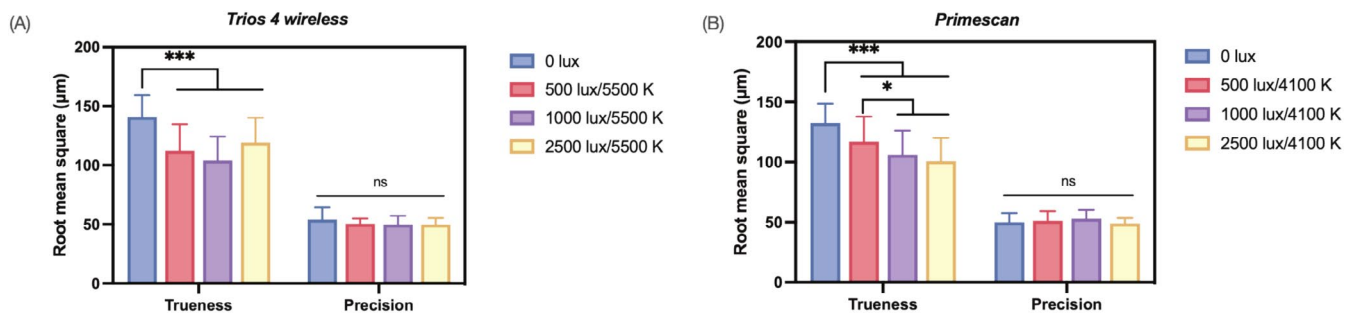


FIGURE 5 | Scanning accuracy (trueness and precision) under various ambient conditions: (A) Trios 4 wireless; (B) Primescan.

participants were aged 18–25 years. Because tooth color tends to darken and increase in chroma with age [53, 54], future studies should include older age groups and a more diverse range of baseline shades to improve generalizability. Additionally, future clinical trials are needed to assess the final esthetic outcomes of actual restorations fabricated based on IOS-derived shade data under controlled operatory lighting. Second, this scanner-head-mounted configuration creates a controlled experimental lighting environment rather than faithfully reproducing routine free-field operatory conditions, thereby partially limiting the clinical generalizability of our findings. The technical specifications of the IOSs' internal light sources are also undisclosed, and parameters such as color temperature, color rendering index, and intensity may interact with ambient lighting in ways that could not be characterized here. In addition, tooth-surface illuminance, which cannot be directly measured, inevitably varies with the dynamic tip-to-tooth distance and angulation during scanning, which may introduce minor discrepancies between the actual tooth-surface illumination and the preset experimental lighting conditions. A custom-designed scanning guide that maintains a constant scan distance could mitigate this variability in future studies. Finally, evaluation of IOS shade matching was constrained to categorical agreement because the tested software does not output raw continuous color coordinates (e.g., CIE $L^*a^*b^*$). Converting categorical shade tabs into continuous coordinates via published tables is methodologically inappropriate for rigorous colorimetric analysis, such as computing color differences against established 50%:50% acceptability thresholds; thus, quantitative assessment was not feasible [55]. To address this limitation and allow for objective evaluation of color differences, developers should enable IOS software to export raw standardized color coordinates in future iterations, thereby enhancing both interpretability and clinical applicability.

5 | Conclusion

Within the limitations of the present study, the following conclusions can be drawn:

1. Ambient light conditions significantly influenced both shade selection capability and scanning trueness of the tested IOSs.
2. Optimal performance was scanner-dependent under clinically relevant lighting sources: TR achieved optimal performance under examination-room lighting (1000 lx) with

a daylight-like source (5500 K), whereas PR performed best under typical dental-operatory lighting (2500 lx, 4100 K).

3. Across the tested lighting conditions, IOS-based shade assignment yielded superior shade matching outcomes when evaluated using the VC shade system.

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Ethics Statement

Ethical approval was obtained from the Institutional Review Board of the local University (approval number: 2022-60).

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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